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ESTIMATION OF PARAMETERS.

Statistical Inference may be divided into 2 major areas: parameter estimation & hypothesis testing. As an example of a parameter estimation problem, suppose that a structural engineer is analysing the tensile strength of a component used in an automobile chasis. Since variability in tensile strength is naturally present between the individual components because of differences in raw material batches, manufacturing processes & measurement procedures, the engineer is interested in estimating the mean tensile strength of the components. In practice, the engineer will use sample data to compute a no that is in some sense a reasonable value (or guess) of the true mean. This number is called a point estimate.

Now consider a situation in which two different reaction temperatures can be used in a chemical process say t_1 and t_2. The engineer conjectures that t_1 results in higher yields than does t_2. Statistical hypothesis testing is a framework for solving problems of this kind.

In inference problems, it is convenient to have a general symbol to rep. the parameter of interest. A Greek symbol (θ) theta is used to represent the parameter.

The objective of point estimation is to select a single number, based on sample data that is most plausible value for θ. A numerical value of a sample statistic will be used as a point estimate.

In general, if X is a random variable with probability distribution $f(x)$, characterized by the unknown

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parameter θ and if $X_1, X_2 \ldots X_n$ is a random sample of size n from X, the statistic $\hat{\Theta}$. The statistic $\hat{\Theta}$ is called the point estimator.

Estimation problems occur frequently in engineering. We often need to estimate

* The mean μ of a single population.
* The variance σ^2 (or std. devn. σ) of a single population.
* The proportion 'p' of items in a population that belong to a class of interest.
* The difference in means of 2 populations, $\mu_1 - \mu_2$
* The difference in two population proportions $p_1 - p_2$.

Reasonable point estimates of these parameters are as follows:

* For μ the estimate $\hat{\mu} = \bar{X}$ the sample mean.
* For σ^2, the estimate is $\hat{\sigma}^2 = s^2$ the sample variance.
* For p, the estimate is $\hat{p} = x/n$ the sample proportion, where x is the no. of items in a random sample of size n that belong to the class of interest.
* For $\mu_1 - \mu_2$, the estimate $\hat{\mu}_1 - \hat{\mu}_2 = \bar{x}_1 - \bar{x}_2$, the diff b/w the sample means of 2 independent random samples.
* For $p_1 - p_2$, the estimate is $\hat{P}_1 - \hat{P}_2$, the diff b/w 2 sample proportions computed from 2 independent random samples.

Sampling distributions and the central limit theorem.

Statistical inference is concerned with making decisions about a population based on the information contained in a random sample from that population.

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Random sample – The random variables $X_1, X_2 \cdots X_n$ are a random sample of size n if (a) the X_i's are independent random variables and (b) every X_i has the same probability distribution.
(This property is known as independent & identically distributed, iid)

The observed data is also referred to as a random sample, but the use of same phrase should not cause any confusion.

Statistic – A statistic is any function of the observations in a random sample.

Sampling Distribution – The probability distribution of a statistic is called sampling distn

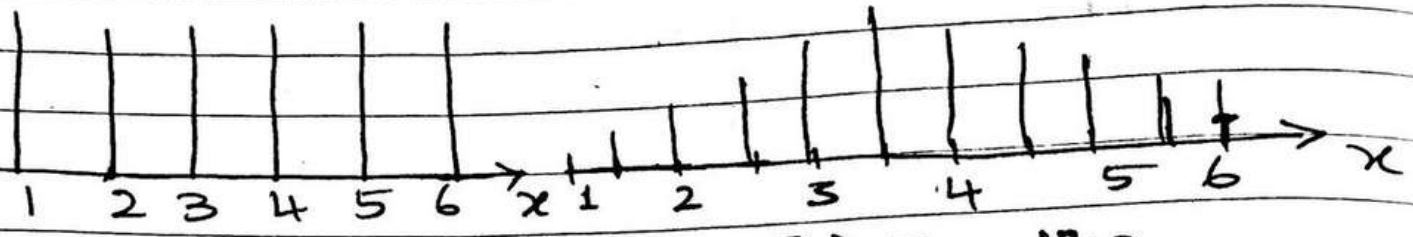
Central Limit Theorem

If $X_1, X_2 \ldots X_n$ is a random sample of size n taken from a population (either finite or infinite) with mean μ and finite variance σ^2, and if $\bar{X}$ is the sample mean, the limiting form of the distribution of

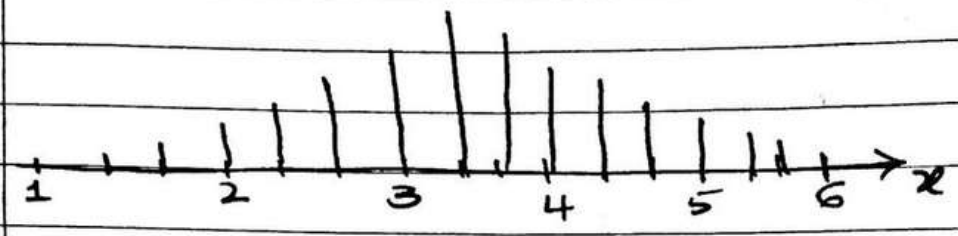
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$ as $n \to \infty$ is the standard normal distribution.

The normal approximation for $\bar{X}$ depends on the sample size n. Fig 1(a) shows the distn. obtained for throws of a single, six sided true die. The probabilities are equal 1/6 for all values. Fig 1(b) shows the distn of avg score obtained when tossing 2 dice. & 1(c), 1(d) & 1(e) shows the distn of avg scores obtained when tossing 3, 5 & 10 dice resp. In case of one die the popln is relatively

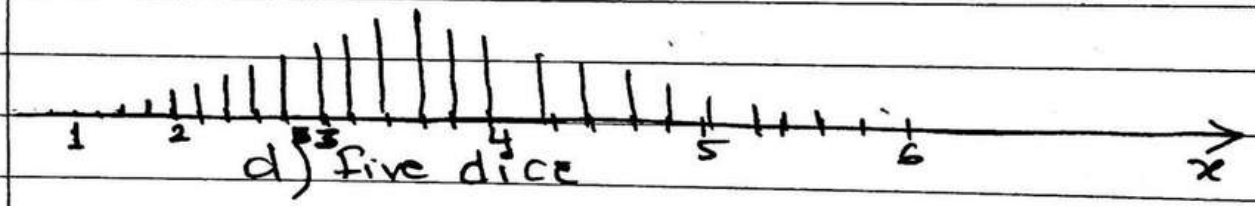
far from normal, distn. of avg is approx. reasonably well by the normal distn for sample sizes as small as 5.



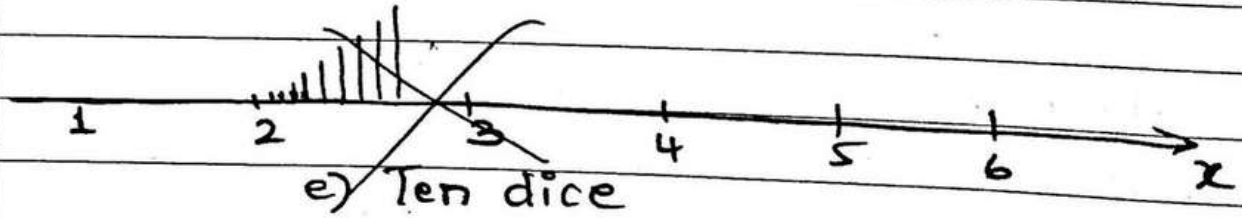
(a) One die. (b) Two dice.



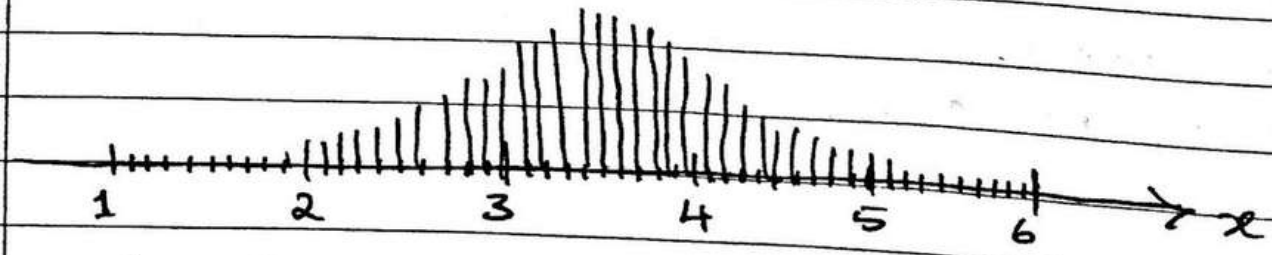
(c) Three dice



d) Five dice



e) Ten dice



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Although the central limit theorem will work well for small sample ($n=4,5$) in most cases, particularly where the population is continuous, unimodal and symmetric, larger sample is required in other situations, depending on the shape of the population. In many cases, if $n \geq 30$, the normal approximation will be satisfactory regardless of the shape of the population. If $n<30$, the central limit theorem will work if the distribution of the population is not severely nonnormal.

General concepts of Point Estimation

Unbiased estimators:- Formally we say that $\hat{\Theta}$ is an unbiased estimator of Θ if the expected value of $\hat{\Theta}$ is equal to Θ.

The point estimator $\hat{\Theta}$ is an unbiased estimator for the parameter θ if

$$E(\hat{\Theta}) = \Theta$$

If the estimator is not unbiased then the difference

$$E(\hat{\Theta}) - \theta \ldots\ldots\ldots$$

is called the bias of the estimator $\hat{\Theta}$

When an estimator is unbiased, the bias is zero:

i.e $E(\hat{\Theta}) - \theta = 0$.

Suppose that X is a random variable with mean μ & variance σ^2. Let $X_1, X_2 \ldots X_n$ be a random sample of size n from the population represented by X. Show that the sample mean $\bar{X}$ and the sample variance S^2 are unbiased estimators of μ & σ^2 respectively.

Consider the sample mean.

We know that $E(\bar{X}) = \mu$

∴ The sample mean $\bar{X}$ is the unbiased estimator of the population mean μ.

For sample variance

$$E(S^2) = E\left[\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}\right] = \frac{1}{n-1} E\sum_{i=1}^{n}(X_i - \bar{X})^2$$

$$= \frac{1}{n-1} E\sum_{i=1}^{n}(X_i^2 + \bar{X}^2 - 2\bar{X}X_i)$$

~~$= \frac{1}{n-1} E\left(\sum_{i=1}^{n} X_i^2 - n\bar{X}^2\right)$~~

$$= \frac{1}{n-1} E\left[\sum(X_i^2) - \sum 2\bar{X}X_i + \sum \bar{X}^2\right)$$

$$= \frac{1}{n-1} E\left[\sum X_i^2 - 2\bar{X}\sum X_i + n\bar{X}^2\right]$$

$$= \frac{1}{n-1} E\left[\sum X_i^2 - 2\bar{X}n\bar{X} + n\bar{X}^2\right] \qquad \bar{X} = \frac{\sum X_i}{n}$$

$$= \frac{1}{n-1} E\left[\sum X_i^2 - 2n\bar{X}^2 + n\bar{X}^2\right]$$

$$= \frac{1}{n-1} E\left[\sum X_i^2 - n\bar{X}^2\right]$$

$$E(S^2) = \frac{1}{n-1}\left[\sum_{i=1}^{n} E(X_i^2) - nE(\bar{X}^2)\right] \quad \text{—— (1)}$$

$E(\sum X_i) = \sum E(X_i)$

$E(cX) = cE(X)$

$Var(X) = E(X^2) - [E(X)]^2$

$E(X^2) = \sigma^2 + \mu^2$

$Var(\bar{X}) = E(\bar{X}^2) - [E(\bar{X})]^2$

$E(\bar{X}^2) = \frac{\sigma^2}{n} + \mu^2$

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$E(X_i^2) = \mu^2 + \sigma^2$ and $E(\bar{X}^2) = \mu^2 + \sigma^2/n$

from eqn (1)

$$E(S^2) = \frac{1}{n-1}\left[\sum_{i=1}^{n} \mu^2 + \sigma^2\right] - n\left[\mu^2 + \frac{\sigma^2}{n}\right]$$

$$= \frac{1}{n-1}\left[n\mu^2 + n\sigma^2\right] - n\mu^2 - \sigma^2$$

$$= \frac{1}{n-1}\left[n\mu^2 + n\sigma^2 - n\mu^2 - \sigma^2\right]$$

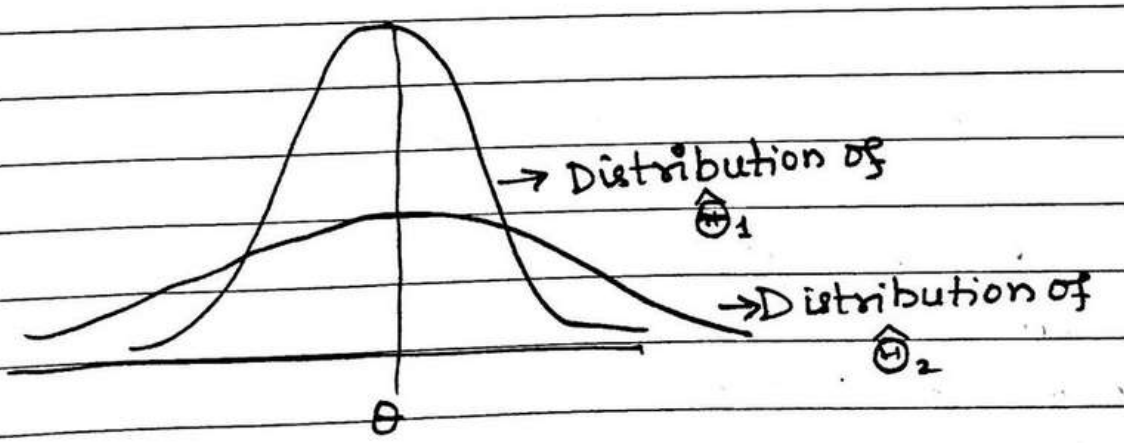
$$= \frac{\sigma^2[n-1]}{n-1} = \sigma^2.$$

∴ the sample variance S^2 is an unbiased estimator of population variance σ^2.

Although S^2 is unbiased for σ^2, S is a biased estimator of σ. For large samples the bias is very small.

Variance of a point estimator.

Suppose there are 2 unbiased estimators of θ. This indicates that the distribution of each distributor is centered at the true value of θ. However the variance of these distributions are different.



A logical principle of estimation, when selecting among several estimators, is to chose the estimator that has min variance.

Min. Variance Unbiased estimator → If we consider all unbiased estimators of θ the one with the smallest variance is called the minimum variance unbiased estimator (MVUE)

If $X_1, X_2 \ldots X_n$ is a random sample of size n from a normal distribution with mean μ and variance σ^2, the sample mean $\bar{X}$ is the MVUE for μ.

Standard Error of an estimator

The standard error of an estimator $\hat{\Theta}$ is its std. dev. given by $\sigma_{\hat{\Theta}} = \sqrt{V(\hat{\Theta})}$. If the standard error involves unknown parameters that can be estimated, substitution of those values into $\sigma_{\hat{\Theta}}$ produced an estimated standard error, denoted by $\hat{\sigma}_{\hat{\Theta}}$

Suppose we are sampling from a normal distribution with mean μ and variance σ^2. Now the distribution of $\bar{X}$ is normal with mean μ & variance σ^2/n, so the standard error of $\bar{X}$ is:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}.$$

In cases in which point estimator is not normally distributed, we can state that so long as the estimator is unbiased, the estimate of the parameter will deviate from the true value by as much as four std errors at most 6% of the time. This is called Chebyshev's inequality.

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1. Det. the bias of the estimator $\hat{\mu} = \bar{X}$ of the parameter μ from a Poi(μ) distribution

$$\text{bias}(\hat{\mu}) = E(\hat{\mu}) - \mu$$
$$= E(\bar{X}) - \mu$$
$$= E\left(\frac{1}{n}\Sigma X_i\right) - \mu.$$
$$= \frac{1}{n}\Sigma E(X_i) - \mu.$$
$$= \frac{1}{n}\Sigma\mu - \mu.$$
$$= \frac{1}{n} n\mu - \mu = \mu - \mu = 0.$$

Estimator is unbiased.

2. Determine the bias of the estimator $\hat{p} = X/n$ of the parameter p from a Bio(n,p) distribution.

$$\text{bias}(\hat{p}) = E(\hat{p}) - p.$$
$$= E\left(\frac{X}{n}\right) - p \qquad X \sim \text{Bin}(n,p)$$
$$E(X) = np$$
$$= \frac{1}{n}E(X) - p$$
$$= \frac{1}{n}np - p$$
$$= p - p = 0.$$

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Mean squared error of an estimator

The mean squared error of an estimator of the parameter θ is defined as:

$$MSE(\hat{\theta}) = E(\hat{\theta} - \theta)^2.$$

It can be rewritten as

$$MSE(\hat{\theta}) = E[\hat{\theta} - E(\hat{\theta})]^2 + [\theta - E(\hat{\theta})]^2$$

$$= V(\hat{\theta}) + (\text{Bias})^2.$$

i.e MSE is equal to variance of the estimator plus the squared bias.

The Mean squared error is an important criterion for comparing 2 estimators. Consider 2 estimators $\hat{\theta}_1$ & $\hat{\theta}_2$ of the parameter θ, and let $MSE(\hat{\theta}_1)$ & $MSE(\hat{\theta}_2)$ be the mean squared error of $\hat{\theta}_1$ & $\hat{\theta}_2$. Then the relative efficiency of $\hat{\theta}_2$ to $\hat{\theta}_1$ is defined as

$$\frac{MSE(\hat{\theta}_1)}{MSE(\hat{\theta}_2)}$$

If this relative efficiency is less than 1, we would conclude $\hat{\theta}_1$ is a more efficient estimation of θ than $\hat{\theta}_2$

Sometimes we find that biased estimators are preferable to unbiased estimators ∵ they have smaller mean squared error. i.e we may be able to reduce the variance of the estimator considerably by introducing a relatively small amount of bias.

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An estimator that has a mean squared error that is less than or equal to the mean squared error of any other estimator, for all values of the parameter θ, is called optimal estimator of θ.

Problems.

1. Suppose we have a random sample of size $2n$ from a population denoted by X, and $E(X) = \mu$ and $V(X) = \sigma^2$. Let

$$\bar{X}_1 = \frac{1}{2n}\sum_{i=1}^{2n} X_i \quad \text{and} \quad \bar{X}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i$$

be two estimators of μ. Which is a better estimate of μ? Explain.

Soln. Determining expected value of both estimators

$$E(\bar{X}_1) = E\left(\frac{1}{2n}\sum_{i=1}^{2n} X_i\right) = \frac{1}{2n}\sum_{i=1}^{2n} E(X_i)$$

$$= \frac{1}{2n}\sum_{i=1}^{2n} \mu = \frac{1}{2n}(2n)\mu = \mu.$$

$$E(\bar{X}_2) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i)$$

$$= \frac{1}{n}\sum_{i=1}^{n} \mu = \frac{1}{n}(n)\mu = \mu.$$

We note that both estimators have same expected value μ.

The best estimator is then the estimator with smallest variance.

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$$V(\bar{X}_1) = V\left(\frac{1}{2n}\sum_{i=1}^{2n} X_i\right) = \left(\frac{1}{2n}\right)^2 \sum_{i=1}^{2n} V(X_i)$$

$$= \frac{1}{4n^2}(2n)\sigma^2 = \frac{\sigma^2}{2n}$$

$$V(\bar{X}_2) = V\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \left(\frac{1}{n}\right)^2 \sum_{i=1}^{n} V(X_i)$$

$$= \frac{1}{n^2} * n * \sigma^2 = \frac{\sigma^2}{n}.$$

We note that the variance of $\bar{X}_1$ is smaller and thus $\bar{X}_1$ is a better estimator.

2. Let $X_1, X_2, \ldots X_7$ denote a random sample from a population having mean μ and variance σ^2. Consider the following fo estimators of μ

$$\hat{\theta}_1 = \frac{X_1 + X_2 + \cdots + X_7}{7}$$

$$\hat{\theta}_2 = \frac{2X_1 - X_6 + X_4}{2}$$

a) Is either estimator unbiased?
b) Which estimator is best? In what sense it is best? Calculale the relative efficiency of the two estimators.

$$\hat{\theta}_1 = \frac{X_1 + X_2 + \cdots + X_7}{7} \qquad \hat{\theta}_2 = \frac{2X_1 - X_6 + X_4}{2}$$

$$E(\hat{\theta}_1) = E\left(\frac{X_1 + X_2 + \cdots + X_7}{7}\right)$$

$$= \frac{1}{7}E(X_1) + \frac{1}{7} * E(X_2) + \frac{1}{7} - \cdots + \frac{1}{7}E(X_7)$$

$$= \frac{1}{7}\mu + \frac{1}{7}\mu + \cdots + \frac{1}{7}\mu = \mu$$

$$E(\hat{\theta}_2) = E\left(\frac{2x_1 - x_6 + x_4}{2}\right)$$

$$= E(x_1) - \frac{1}{2}E(x_6) + \frac{1}{2}E(x_4)$$

$$= \mu - \frac{1}{2}*\mu + \frac{1}{2}\mu = \mu.$$

Hence $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators of μ

ii. Let us determine the variances of estimators

$$Var(\hat{\theta}_1) = Var\left[\frac{x_1 + x_2 + \cdots - x_6 + x_7}{7}\right]$$

$$= \frac{1}{49}Var(x_1) + \frac{1}{49}Var(x_2) + \cdots + \frac{1}{49}Var(x_7)$$

$$= \frac{1}{49}\sigma^2 + \frac{1}{49}\sigma^2 + \frac{1}{49}\sigma^2 + \frac{1}{49}\sigma^2 + \frac{1}{49}\sigma^2 + \frac{1}{49}\sigma^2 + \frac{1}{49}\sigma^2$$

$$= \frac{1}{7}\sigma^2.$$

$$Var(\hat{\theta}_2) = Var\left[\frac{2x_1 - x_6 + x_4}{2}\right]$$

$$= \frac{1}{4}Var[2x_1] + \frac{1}{4} * Var[x_6] + \frac{1}{4}Var[x_4]$$

$$= \frac{4}{4}Var[x_1] + (+)\frac{1}{4}Var[x_6] + \frac{1}{4}Var[x_4]$$

$$= \sigma^2 + \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2$$

$$= \frac{3}{2}\sigma^2.$$

The best estimator is the one with a smaller variance

$\therefore \hat{\theta}_1$ is the best estimator.

Relative efficiency of two estimators

$$= \frac{3/25^2}{1/75^2} = \frac{21}{2}.$$

3. Suppose that $\hat{\theta}_1$ & $\hat{\theta}_2$ are estimators of the parameter θ. We know that $E(\hat{\theta}_1) = \theta$ and $E(\hat{\theta}_2) = \theta/2$ $V(\hat{\theta}_1) = 10$, $V(\hat{\theta}_2) = 4$. Which estimator is best? In what sense is it best.

Since mean of both estimators vary $\Rightarrow$ mean sq. error of both estimators also varies.

$$MSE(\hat{\theta}_1) = V(\hat{\theta}_1) + [Bias(\hat{\theta}_1)]^2$$
$$= 10 + [E(\hat{\theta}_1) - \theta]^2$$
$$= 10 + [\theta - \theta]^2 = \frac{10}{2}$$
$$MSE(\hat{\theta}_2) = 4 + [E(\hat{\theta}_2) - \theta]^2$$
$$= 4 + \left[\frac{\theta}{2} - \theta\right]^2$$
$$= 4 + \frac{\theta^2}{4}$$

$$\text{Relative efficiency} = \frac{MSE(\hat{\theta}_1)}{MSE(\hat{\theta}_2)} = \frac{V(\hat{\theta}_1)}{V(\hat{\theta}_2)}$$

$$= \frac{10}{4 + \frac{\theta^2}{4}} = \frac{40}{16 + \theta^2}.$$

If $\theta^2 = 24$, relative efficiency is $1 \Rightarrow \hat{\theta}_1$ & $\hat{\theta}_2$ are equally efficient.

If $\theta^2 > 24$, relative efficiency is $< 1 \Rightarrow \hat{\theta}_1$ is more efficient

If $\theta^2 < 24$, relative efficiency is $> 1 \Rightarrow \hat{\theta}_2$ is more efficient

4. Data on pull-off force (pounds) for connectors used in an automobile engine application are as follows-
79.3, 75.1, 78.2, 74.1, 73.9, 75.0, 77.6, 77.3, 73.8, 74.6, 75.5, 74.0, 74.7, 75.9, 72.9, 73.8, 74.2, 78.1, 75.4, 76.3

75.3, 76.2, 74.9, 78.0, 75.1, 76.8

a) Calculate a point estimate of the mean pull-off force of all connectors in the population. State which estimator you used and why?

b) Calculate a point estimate of the pull-off force value that separates the weakest 50% of the connectors in the population from the strongest 50%.

c) Calculate point estimates of the population variance & the population standard deviation.

d) Calculate the std. error of the point estimate found in part (a).

e) Calculate a point estimate of the proportion of all connectors in the population whose pull-off force is less than 73 pounds.

(a) Sample Mean $\bar{X} = 1966/26 = 75.62$ pounds

Population Mean (μ) = Sample Mean ($\bar{x}$) = 75.62 pounds

It is known that sample mean and variance are unbiased estimators of population.

b) We have to find Median of the sample

72.9, 73.8, 73.8, 73.9, 74.0, 74.1, 74.2, 74.6, 74.7, 74.9, 75.0, 75.1, 75.1, 75.3, 75.4, 75.5, 75.9, 76.2, 76.3, 76.8, 77.3, 77.6, 78.0, 78.1, 78.2, 79.3

Sample Median = 75.2 pounds = Population median

c) Good pt. estimates of popln. variance & popln std. variance.

Population variance (σ^2) $\cong$ Sample Variance (s^2)
$= 2.7382$ pounds^2

Population std. devn = sample std. devn $= 1.6547$ pounds

d) The standard error of the calculated point estimate for the mean pull-off force is

$$SE(\bar{x}) = \hat{\sigma}_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{1.6547}{\sqrt{26}} = 0.3245.$$

The std error is about 0.4% of the sample mean, $\Rightarrow$ that we have obtained a relatively precise pt. estimate of pull-off force for this case. If we can assume that pull-off force is normally distributed then:

$$2\hat{\sigma}_{\bar{x}} = 2(0.3245) = 0.6490$$

$\Rightarrow$ that we are confident that the true mean-pull off force for this case is within the interval 75.62 ± 0.649 or $(74.97, 76.26)$ pounds.

e) A point estimate of the proportion of all connectors in the population whose pull-off force < 73 pounds is:

$$\hat{p} = \frac{X}{n} = \frac{\#\text{ of specimens less than 73 pounds}}{\text{Total \# of specimens.}}$$

$$= \frac{1}{26} = 0.038462.$$

Methods of Point Estimation.

Method of Moments.

Moment.

Let $X_1, X_2, \ldots, X_n$ be a random sample from the probability distribution $f(x)$, where $f(x)$ can be a discrete probability mass function or a continuous prob. density function. The kth population moment (or distribution moment) is $E(X^k)$, $k = 1, 2, \ldots$. The corresponding k^{th} sample

moment is $1/n \sum_{i=1}^{n} X_i^k$, $k = 1, 2 \ldots$

Moment Estimators.

Let $X_1, X_2 \ldots X_n$ be a random sample from either a probability mass function or probability density function with m unknown parameters $\theta_1, \theta_2 \ldots, \theta_m$. The moment estimators $\hat{\theta}_1, \hat{\theta}_2, \ldots \hat{\theta}_m$ are found by equating the first m population moments to the first m sample moments and solving the resulting equations for the unknown parameters.

2. Method of Maximum Likelihood.

Maximum Likelihood estimator – Suppose that X is a random variable with prob. distribution $f(x:\theta)$, where θ is a single unknown parameter. Let $x_1, x_2 \ldots, x_n$ be the observed values in a random sample of size n. Then the likelihood function of the sample is

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \cdots \cdot f(x_n : \theta)$$

Note that the likelihood function is now a function of only the unknown parameter θ. The maximum likelihood estimator (MLE) of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

Problems.

Propertie of a Maximum likelihood pstimator

Under very general & not restrictive conditions, when the sample size n is large and if $\hat{\theta}$ is the max. likelihood estimator of the parameta θ,

(1) $\hat{\theta}$ is an approximately unbiased estimator for θ

$$E(\hat{\theta}) \cong \theta$$

(2) the variance of $\hat{\theta}$ is nearly as small as the variance that could be obtained with any other estimator and

(3) $\hat{\theta}$ has an approximate normal distribution.

Problems:

1. Consider the Poisson distribution with parameter λ. Find the maximum likelihood estimator of λ, based on a random sample of size n

Let random variable X follow poisson distribution with parameter λ

$$f(x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad x = 0, 1, 2 \cdots n.$$

Let us find the max. likelihood estimator of λ, based on random sample of size n

$$L(x, \lambda) = \frac{e^{-\lambda}\lambda^{x_1}}{x_1!} \cdot \frac{e^{-\lambda}\lambda^{x_2}}{x_2!} \cdots \frac{e^{-\lambda}\lambda^{x_n}}{x_n!}$$

$$L(x, \lambda) = \prod_{i=1}^{n} \frac{e^{-n\lambda}\lambda^{x_i}}{x_i!}$$

Let us apply logarithm on both sides

$$\log L(\lambda) = \log\left(\prod_{i=1}^{n} \frac{e^{-n\lambda}\lambda^{x_i}}{x_i!}\right)$$

$$= -n\lambda + \log\left(\frac{\lambda^{\sum_{i=0}^{n} x_i}}{\prod_{i=0}^{n} x_i!}\right)$$

$$= -n\lambda + \sum_{i=0}^{n} x_i \log\lambda - \log(x!)$$

Differentiating w.r.t λ & equate to zero

$$\frac{d \log L(\lambda)}{d\lambda} = -n + \sum_{i=0}^{n} \frac{x_i}{\lambda} + 0 = 0$$

classmate

$$\Rightarrow \sum_{i=0}^{n} \frac{x_i}{\lambda} = n$$

$$\Rightarrow \boxed{\lambda = \frac{\sum_{i=0}^{n} x_i}{n}} \Rightarrow \boxed{\lambda = \bar{x}}$$

2. Let X be a geometric random variable with parameter p. Find the max. likelihood estimator of p, based on a random sample of size n.

X is a geometric random variable with parameter p & its density function:

$$f(x) = (1-p)^{x-1} * p \quad ; \quad x = 1, 2 \text{ - - -}$$

Max likelihood estimator of p based on random sample of size n

$$L(p) = p*(1-p)^{x_1-1} * p*(1-p)^{x_2-1} \cdots p(1-p)^{x_n-1}$$

$$= p^n \prod_{i=1}^{n} (1-p)^{x_i-1}.$$

Now, apply log on both sides

$$\log L(p) = \log p^n + \log (1-p)^{\sum_{i=1}^{n} x_i - n}$$

$$= n \log p + \left(\sum_{i=1}^{n} x_i - n\right) \log (1-p)$$

Differentiate w.r.t p & equate to zero

$$\frac{d \log L(p)}{dp} = \frac{n}{p} + \frac{\left(\sum_{i=1}^{n} x_i - n\right)}{1-p} (-1) = 0$$

$$\frac{n}{p} = \left(\sum_{i=1}^{n} x_i - n\right) \Rightarrow \frac{1}{\bar{x}} - 1 = \sum_{i=1}^{n} x_i - n$$

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$$\frac{1}{p} = 1 + \sum_{i=1}^{n} \frac{x_i - n}{n}$$

$$\frac{1}{p} = \sum_{i=1}^{n} \frac{x_i - n + n}{n}$$

$$\boxed{\hat{p} = \frac{n}{\sum_{i=1}^{n} x_i}}$$

3. Let X be a random variable with the foll prob. distn.

$$f(x) = \begin{cases} (\theta+1)\, x^{\theta}, & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Find the max. likelihood estimator of θ, based on a random sample of size n.

Let $x_1, x_2, x_3 \cdots x_n$ be random sample

The likelihood function $(\theta+1) * x_1^{\theta} * (\theta+1) * x_2^{\theta} * \cdots (\theta+1) x_n^{\theta}$.

$$L(\theta) = \prod_{i=1}^{n} f(x_i) = (\theta+1)^n [x_1 * x_2 * x_3 * \cdots x_n]^{\theta}$$

Taking log on b.t.s & differentiating it w.r.t θ & equate to zero.

$$\frac{d}{d\theta} \log(L(\theta)) = \frac{d}{d\theta} [n \log(\theta+1) + \theta \log(x_1 x_2 x_3 \cdots x_n)] = 0.$$

$$= \frac{n}{\theta+1} + \log(x_1 x_2 x_3 \cdots x_n) = 0.$$

$$= \frac{n}{\theta+1} + \sum_{i=1}^{n} \log(x_i) = 0.$$

$$\Rightarrow \sum_{i=1}^{n} \log(x_i) = \frac{-n}{\theta+1}$$

$$\Rightarrow \theta+1 = \frac{-n}{\sum_{i=1}^{n} \log(x_i)} \qquad \Rightarrow \theta = -\left[1 + \frac{n}{\sum_{i=1}^{n} \log(x_i)}\right]$$

4. 10 values collected at random from a Poisson distribution with mean μ are: 10, 11, 15, 17, 9, 13, 18, 11, 15, 16. Estimate the mean parameter μ using method of moments.

What significance does the mean square error hold for an estimator? How is it used to evaluate an estimator?

Soln. x_i - $i = 1, 2, \ldots 10$ $\quad X \sim Poi(\mu)$

We know $E(\bar{X}) = \mu$

Method of moments: $E(X) = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$.

$$= \frac{1}{10}\sum_{i=1}^{10} x_i$$

$$\therefore \hat{\mu} = \frac{1}{10}\sum x_i = \frac{135}{10} = 13.5.$$

MSE combines effect of bias and variance of estimator, smaller the MSE, the better is estimator

$$MSE = bias^2 + Var.$$

It is possible that out of 2 estimators one with higher bias or variance may have lower MSE.

5. The random variable X represents the no. of defective parts in a lot of 90 parts. A sample of 25 lots give x_i, $i = 1, 2, \ldots 25$ where each x_i = no of defective parts in i^{th} lot. Derive the method of moments estimator for the proportion of a part being defective

$X \sim$ Binomial $(90, p)$ where p = proportion defecti

Method of moments : $E(X)$ = Sample moment

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$$90p = \frac{1}{25}\sum_{i=1}^{25} x_i .$$

$$\Rightarrow \hat{p} = \frac{1}{90} * \left[\frac{1}{25}\sum_{i=1}^{25} x_i\right] = \frac{\bar{x}}{90}.$$

Population mean (mp) for binomial can be estimated as

$$\bar{x},\ \frac{\max(x_i) + \min(x_i)}{2},\ \bar{y}$$

where $\bar{y}$ is calculated Σy_i y_i are 23 values excluding $\min(x_i)$ & $\max(x_i)$

$$\text{MSE of } (\bar{x}) = [\text{Bias}(\bar{x})]^2 + \text{Var}(\bar{x})$$
$$= 0 + \frac{90p(1-p)}{25}$$

$$\text{MSE}\left(\frac{\max(x_i)+\min(x_i)}{2}\right) = 0 + \frac{90p(1-p)}{2}.$$

$$\text{MSE}(\bar{y}) = 0 + \frac{90p(1-p)}{23}$$

$\therefore \bar{x}$ is best, since its MSE is minimum

[Bias is used for unbiasedness.
Std. error used to det. low spread.
MSE — used for checking consistency.]

Normal distribution Moment Estimator

Suppose that $x_1, x_2, \ldots x_n$ is a random sample from a normal dist with parameters μ & σ^2.

For normal dist $E(X) = \mu$ & $E(X^2) = \mu^2 + \sigma^2$.

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$E(X-\mu)=0$
$E[(X-\mu)^2]=\sigma^2$.

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Equating $E(X)=\bar{X}$ and $E(X^2)$ to $\frac{1}{n}\sum_{i=1}^{n} x_i^2$ gives

$$\mu=\bar{X},\quad \mu^2+\sigma^2=\frac{1}{n}\sum_{i=1}^{n} x_i^2.$$

Solving these equations

$$\hat{\mu}=\bar{X},\quad \hat{\sigma}^2=\frac{1}{n}\sum_{i=1}^{n} x_i^2-\hat{\mu}^2$$

$$\hat{\sigma}^2=\sum_{i=1}^{n} x_i^2-\frac{n*\left(\frac{1}{n}*\sum_{i=1}^{n} x_i\right)^2}{n}$$

$$=\sum_{i=1}^{n} x_i^2-\frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}.$$

$$=\frac{1}{n}\sum_{i=1}^{n} x_i^2-\bar{X}^2=\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{X})^2$$

$E[X-\mu]=0 \Rightarrow \frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})=0$. — (i)

$E[(X-\mu)^2]=\sigma^2$.

$\hookrightarrow \frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^2=\sigma^2$. — (ii)

Solving

$\hat{\mu}=\frac{1}{n}\sum_{i=1}^{n} x_i \Rightarrow \boxed{\hat{\mu}=\bar{X}}$

$$\hat{\sigma}^2=\frac{1}{N}\sum_{i=1}^{n}(x_i-\hat{\mu})^2=\frac{1}{N}\sum_{i=1}^{n}(x_i-\bar{X})^2$$

Notice that the moment estimator of σ^2 is not an unbiased estimator.

7. Consider the exponential distribution with parameter λ. Derive the likelihood function of random sample of size n.

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The likelihood fn of exponential distn is

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_i} \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}$$

$$\ln L(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^{n} x_i$$

Now $$\frac{d \ln L(\lambda)}{d\lambda} = \frac{n}{\lambda} - 1 \sum_{i=1}^{n} x_i$$

Equating this to zero we get.

$$\hat{\lambda} \quad \frac{n}{\lambda} - 1 \sum_{i=1}^{n} x_i = 0.$$

$$\text{or} \quad \frac{n}{\lambda} = \sum_{i=1}^{n} x_i.$$

$$\text{or} \quad \hat{\lambda} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}}.$$

$$\boxed{\hat{\lambda} = \frac{1}{\bar{x}}}$$

Thus the max. likelihood estimator of λ is the reciprocal of the sample mean.

8. Consider the probability density function

$$f(x) = c(1+\theta x) \quad -1 \le x \le 1$$

a) Find the value of the constant c.
b) What is the moment estimator for θ?
c) Show that $\hat{\theta} = 3\bar{X}$ is an unbiased estimator for θ.
d) Find the max. likelihood estimator for θ.

Soln A value prob. density function satisfies

$$\int_x f(x)\,dx = 1.$$

$$\therefore \int_{-1}^{1} c(1+\theta x)\,dx = 1.$$

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$$c*\left(x+\frac{\theta x^2}{2}\right)\Big|_{-1}^{1} = 1.$$

$$= c*(1+\frac{\theta}{2}) - (c*(-1+\frac{\theta}{2})) = 1.$$

$$\Rightarrow c + \frac{c\theta}{2} + c - \frac{c\theta}{2} = 1.$$

$$\Rightarrow 2c = 1 \Rightarrow c = \frac{1}{2}.$$

b. Moment estimator for θ

$$\hat{\mu} = \int_{-1}^{1} x \cdot f(x)\, dx.$$

$$\hat{\mu} = \int_{-1}^{1} x * \frac{1}{2}(1+\theta x)\, dx.$$

$$= \frac{1}{2}\int_{-1}^{1} (x+\theta x^2)\, dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2} + \frac{\theta x^3}{3}\right]_{-1}^{1}.$$

$$\hat{\mu} = \frac{1}{2}\left[\frac{1}{2}(1^2-(-1)^2) + \frac{\theta}{3}[1^3-(-1)^3]\right]$$

$$= \frac{1}{2}\left[0 + \frac{\theta}{3}[1^3+1^3]\right] = \frac{2}{2} * \frac{\theta}{3}$$

$$\boxed{\hat{\mu} = \frac{\theta}{3}}$$

c. Sample estimator of mean is $\bar{X}$, $\hat{\mu} = \frac{\theta}{3}$

$$\hat{\mu} = \frac{\theta}{3} \Rightarrow \theta = 3\hat{\mu} \Rightarrow 3\bar{X}.$$

Hence, moment estimator of θ is an unbiased estimator of $3\bar{X}$.

We have to prove $E(3\bar{X}) = \theta$.

$$3E(\bar{X}) = 3\hat{\mu}$$

(∵ Since sample mean $\bar{x}$ is an unbiased estimator for population mean $\hat{\mu}$)

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So $E(3\bar{x}) = 3\hat{\mu} = 3 * \frac{\theta}{3} = \theta$.

Maximum likelihood estimator for θ

$$f(x) = \frac{1}{2}(1+\theta x)$$

$$L(\theta) = \prod_{i=1}^{n} f(x_i) = \frac{(1+\theta x_1)(1+\theta x_2)(1+\theta x_3) \cdots * (1+\theta x_n)}{2^n}$$

$$L(\theta) = \frac{1}{2^n} \prod_{i=1}^{n} (1+\theta x_i)$$

Taking log.

$$\text{Log } L(\theta) = -n \log 2 + \sum_{i=1}^{n} \log(1+\theta x_i)$$

Now

$$\frac{d}{d\theta} L(x, \theta) = 0 + \sum_{i=1}^{n} x_i * \frac{1}{(1+\theta x_i)}$$

Equating this to zero

$$\sum_{i=1}^{n} \frac{x_i}{(1+\theta x_i)} = 0 \Rightarrow \sum_{i=1}^{n} x_i = 0$$

By observation the value of θ that maximizes the likelihood is $\max(x_i)$

Commonly used estimators are as follows

1. Maximum likelihood estimators
2. Bayes estimators.
3. Method of moments estimators
4. Least squares.
5. Minimum mean squared error (MMSE) or Bayes least squared error (BLSE)
6. Minimum variance unbiased estimator (MVUE)
7. Best Linear unbiased estimator (BLUE)
8. Non linear system Identification.
9. Unbiased estimators.
10. Markov chain Monte Carlo.
11. Particle filter.
12. Kalman filter.
13. Wiener filter

There are several criterias proposed for estimators.

1. Unbiased estimator (long run average of the estimate equals the true parameter value)
2. Consistency estimator (with more & more observations the estimate gets nearer the true parameter value)
3. Efficient estimator (has small variability)
4. Minimum variance among all unbiased estimators

An estimator Y for a parameter θ is unbiased if $E(Y) = \theta$. Intuitively, Y is unbiased if its long-term average value is equal to θ. In normal population model, $\bar{X}$ is an unbiased estimator of μ. This is $\because E(X_i) = \mu$ for each i & hence $E(\bar{X}) = \mu$. Also s^2 is an unbiased estimator of σ^2.

On the other hand, if you put Y as the largest of the observations, then Y is not an unbiased estimator. It appears that, after all, the max is one of the obsvns & each obsvn. has expected value μ so

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Y must have the same property. But it is not so. Indeed if the sample size is at least two, then $E(Y) > \mu$.

An estimator is consistent if it gets closer & closer to the parameter value as the sample size increases. i.e an estimator Y, for a parameter θ is consistent if for any $\epsilon > 0$ we have $P(|Y-\theta| \geq \epsilon) \to 0$ as $n \to \infty$.

For normal population model $\bar{X}$ is a consistent estimator of μ This is because given any $\epsilon > 0$.

$$P(|\bar{X}-\mu| \geq \epsilon) \leq \frac{Var(\bar{X})}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2}\frac{1}{n} \to 0$$

This argument shows that for any model, with μ denoting the popln mean, $\bar{X}$ is a consistent estimator of μ. This depends only on the fact that the variance of $\bar{X}$ is σ^2/n where σ^2 is population variance, assumed to be finite. In fact the same argument gives us a gen. fact: If Y_n is an unbiased estimator (based on a sample of size n) of θ and if $Var(Y_n) \to 0$, then Y_n is a consistent estimator of θ

If Y is unbiased estimator of θ, then, of course, $E(Y-\theta)^2$ is nothing but the variance of Y. However, if Y is not biased, then this is no longer the variance of Y. This quantity $E(Y-\theta)^2$ is called the Mean Square Error (MSE)
An estimator is said to have min. MSE if this quantity is the least possible. When the estimator is unbiased, then MSE being its variance, an unbiased estimator with min MSE is called Min. Variance Unbiased Estimator (MVUE). $\bar{X}$ has min. variance among all (unbiased) estimators which are linear combinations of $X_1, \ldots X_n$.

Method of moments.

Let X: Random variable with density function $f(x:\theta_1,\theta_2)$

Parameters to be estimated θ_1, θ_2.

Have a random sample: $X_1, \ldots, X_n$.

1. Calculate the first two sample moment

$$m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i \qquad m_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2.$$

2. Calculate $E(X)$ and $E(X^2)$ the first two population moments:

$$E(X^k) = \int_{-\infty}^{\infty} x^k f(x:\theta_1,\theta_2)\, dx.$$

The results are in terms of θ_1 & θ_2

3. Solve for θ_1, θ_2 the simultaneous equations.

$$E(X) = m_1, \quad E(X^2) = m_2.$$

The solutions are the method of moments estimators of θ

Eg. Consider a random variable $X \sim N(\mu, \sigma^2)$

$$f(x:\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]}$$

Parameters: μ & σ^2.

Random sample: $X_1 \ldots\ldots X_n$

1. The first 2 sample moments:

$$m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}.$$

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$m_2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2$ = $S_x = \frac{1}{N}\sum_{i=1}^{n}(x_i - \bar{x})^2$

2. The first and second theoretical moments for normal distn are

$\hat{\mu}_1 = E(X) = \mu$.

$Var(x) = \sigma^2$.

$\hat{\mu}_2 = E(X^2) = \mu^2 + \sigma^2$.

3. Solving the equations.

$\mu = \bar{X}$ and $\mu^2 + \sigma^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2$.

We have the method of moment estimate

$\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \bar{X}^2$

$= \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{X})^2$.

2. Suppose that X is a discrete random variable with the foll prob: where $0 \le \theta \le 1$ is a parameter.

X	0	1	2	3
P(X)	$2\theta/3$	$\theta/3$	$2(1-\theta)/3$	$(1-\theta)/3$

The foll. 10 independent observations were taken from such a distn (3, 0, 2, 1, 3, 2, 1, 0, 2, 1). Please use the method of moment to find the estimate of θ.

Soln The theoretical mean value is

$E(X) = \sum_{x=0}^{3} x P(x) = 0 + \frac{2\theta}{3} + \frac{\theta}{3} + 2\cdot\frac{(2-\theta)}{3} + 3\left(\frac{1-\theta}{3}\right)$

$= \frac{7}{3} - 2\theta$.

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The sample mean value is

$$\bar{X} = \sum_{i=1}^{n} \frac{X_i}{n} = \frac{3+0+2+1+3+2+1+0+2+1}{10} = 1.5$$

We need to solve the eqn. $\frac{7}{3} - 2\theta = 1.5$ & we finally get the method of moment estimator $\boxed{\hat{\theta} = \frac{5}{12}}$

$$\text{or } 2\hat{\theta} = \frac{7}{3} - 1.5 \Rightarrow \theta = \frac{7 - 4.5}{3 \times 2} \rightarrow \boxed{\frac{2.5}{6} \text{ or } \frac{5}{12}}$$

3. Use the method of moment to estimate the parameter θ for the uniform density $f(x|\theta) = \frac{1}{\theta}$ with $0 \le x \le \theta$, based on a random sample $X_1 \cdots X_n$.

Soln. Theoretical moment for this distn is

$$E(X) = \int_0^\theta x * \frac{1}{\theta} dx = \frac{x^2}{2\theta}\Big|_0^\theta = \frac{\theta^2}{2\theta} - 0 = \frac{\theta}{2}$$

First sample moment is $\bar{X}$.

Equating both.

$$E(X) = \bar{X} \Rightarrow \frac{\theta}{2} = \bar{X} \Rightarrow \hat{\theta} = 2\bar{X} = 2 * \frac{1}{n} \sum_{i=1}^{n} X_i$$

is the method of moment estimate.

4. For Bernoulli distribution

$X \sim Bern(p)$

$$P_*(x) = p^x (1-p)^{1-x}$$

population Mean

$$E(X) = p.$$

Sample mean

$$\bar{X} = \frac{1}{N}\sum_{i=1}^{n} x_i$$

$$\hat{p} = \bar{X} = \frac{1}{N}\sum_{i=1}^{n} x_i$$

5. Geometric Distribution

$X \sim Geo(p)$

$$P_X(x) = p(1-p)^{x-1}$$

$$E(X) = 1/p \qquad \bar{X} = \frac{1}{N}\sum_{i=1}^{n} x_i$$

$$\hat{p} = \frac{1}{\bar{X}} = \frac{N}{\sum_{i=1}^{n} x_i}$$

6. Poisson Distn

$X \sim Poisson(\lambda)$

$$P_X(\lambda) = \frac{e^{-\lambda}\lambda^x}{x!} \quad x \geq 0$$

$$E(X) = \lambda. \qquad \bar{X} = \frac{1}{N}\sum_{i=1}^{n} x_i$$

$$\hat{\lambda} = \bar{X} = \frac{1}{N}\sum_{i=1}^{n} x_i$$

7. χ^2_p chisquare.

$$f_x(x) = \frac{1}{\Gamma\left(\frac{p}{2}\right)2^{p/2}} x^{p/2-1} e^{-x/2}.$$

$$E(X) = p \qquad \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$\hat{p} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

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Maximum Likelihood Method problems

1 The foll samples 0, 1, 0, 0, 1 0 are from a binomial distribution which has the form $P(X=0) = (1-\mu)$ & $P(X=1) = \mu$

Find the max. likelihood estimate of μ

Soln $L(\mu) = P(X=0)\ P(X=1)\ P(X=0)\ P(X=0)\ P(X=1)\ (P(X=0$

$= (1-\mu)\ \mu\ (1-\mu)\ (1-\mu)\ \mu\ (1-\mu)$

$= (1-\mu)^4 \mu^2.$

$\log L(\mu) = \log(1-\mu^4, \mu^2) = 4\log(1-\mu) + 2\log\mu$

$$\frac{\partial}{\partial u} \log L(\mu) = 4\frac{1}{1-\mu}(-1) + 2\times\frac{1}{\mu} = 0$$

$$-4\mu + 2(1-\mu) = 0.$$

$$\Rightarrow -4\mu + 2 - 2\mu = 0.$$

$$\Rightarrow -6\mu + 2 = 0.$$

$$\mu = \frac{2}{6} = \frac{1}{3}.$$

2nd derivative.

$$\frac{\partial^2}{\partial\mu^2} \log l(u) = \frac{-4}{(1-\mu)^2} + \frac{2}{\mu^2}$$ it is -ve.

hence the max likelihood ~~is~~ estimate is $\boxed{\frac{1}{3}}$.

2. Given.

Samples 5, 9, 3, 12, 14 from a Poisson distribution which has the form

$$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

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What is the max likelihood estimate of lambda

$$L(\lambda) = P(X=5)\ P(X=9)\ P(X=3)\ P(X=12)\ P(X=14)$$

$$= \frac{e^{-\lambda}\lambda^5}{5!} * \frac{e^{-\lambda}\lambda^9}{9!} * \frac{e^{-\lambda}\lambda^3}{3!} * \frac{e^{-\lambda}\lambda^{12}}{12!} * \frac{e^{-\lambda}\lambda^{14}}{14!}$$

$$= \frac{e^{-5\lambda}\lambda^{43}}{5!\,9!\,3!\,12!\,14!}$$

$$\log L(\lambda) = 43\log\lambda - 5\lambda - \log(5!\,9!\,3!\,12!\,14!)$$

$$= \frac{43*1}{\lambda} - 5 = 0.$$

$$\boxed{\Rightarrow \lambda = \frac{43}{5}}$$

2nd derivative

$$\frac{\partial^2}{\partial\lambda^2}\log L(\lambda) = -\frac{43}{\lambda^2}$$ for any λ it is -ve

hence

$$\boxed{\lambda = \frac{43}{5}}$$

Analysis of Statistical Intervals: Single sample

An interval estimate for a population parameter is called a confidence interval. We cannot be certain that the interval contains the true, unknown population parameter — we only use a sample from the full population to compute the point estimate and the interval. However the CI is constructed so that we have confidence that it does contain the unknown population parameter. CIs are widely used in Enginering & the sciences.

Interval estimates can be of three types!

1. A confidence interval bounds popwlation or distribution parameters (such as mean viscosity).
2. A tolerance interval bounds a selected proportion of a distribution.
3. A prediction interval bounds future observations from the population or distribution.

CONFIDENCE INTERVAL ON THE MEAN OF A NORMAL DISTRIBUTION, VARIANCE KNOWN

Suppose that $X_1, X_2 \ldots X_n$ is a random sample from a normal distn. with unknown mean μ & known variance σ^2. We know that the sample mean $\bar{X}$ is normally distributed with mean μ & variance σ^2/n. We may standardize $\bar{X}$ by subtracting the mean & dividing by std. devn. which results in the variable

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

The random variable Z has a std. normal distn

A confidence interval estimate for μ is an interval of the

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form $l \leq \mu \leq u$, where the endpoints 'l' and 'u' are ~~values of random variables~~ computed from sample data. Bcoz different samples will produce diff values of 'l' and 'u', these end points are values of random variables L and 'U' respectively.

Suppose that we can det. values of L and U such that the foll. probability statement is true.

$$P\{L \leq \mu \leq U\} = 1-\alpha$$

where $0 \leq \alpha \leq 1$

There is a probability of $1-\alpha$ of selecting a sample for which the CI will contain the true value of μ. Once we have selected the sample so that $X_1 = x_1, X_2 = x_2 \ldots X_n = x_n$ & computed l and u, the resulting confidence interval for μ is

$$l \leq \mu \leq u$$

The end points or bounds l and u are called the lower & upper confidence limits resp. & $1-\alpha$ is called confidence coefficient.

In our problem situation, because $Z = (\bar{X}-\mu)/(\sigma/\sqrt{n})$ has a standard normal distribution, we may write

$$P\left\{-z_{\alpha/2} \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right\} = 1-\alpha$$

Now manipulate the terms inside brackets by (1) multiplying through $\sigma/\sqrt{n}$ (2) Subtracting $\bar{X}$ from each term & (3) multiplying through by -1. We get

$$P\left\{\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\}$$

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C.I on the mean Variance known

If $\bar{x}$ is the sample mean of a random sample of size n from a normal population with known variance σ^2, a $100(1-\alpha)\%$ CI on μ is given by

$$\bar{x} - z_{\alpha/2}\sigma/\sqrt{n} \leq \mu \leq \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}$$

where $z_{\alpha/2}$ is the upper $100\alpha/2\%$ pt of std. normal distribution

Note: For a fixed sample size n and std. devn. σ, the higher the confidence level, the longer the resulting CI.

The length of confidence interval is a measure of the precision of estimation.

Choice of sample size

If $\bar{x}$ is used as an estimate of μ, we can be $100(1-\alpha)\%$ confident that the error $|\bar{x}-\mu|$ will not exceed a specified amount E when the sample size is

$$n = \frac{(z_{\alpha/2}\sigma)^2}{E}.$$

If the rhs of this eqn is not a integer, it must be rounded up. This will ensure that the level of confidence does not fall below $100(1-\alpha)\%$. Notice that 2E is the length of resulting confidence interval.

One sided Confidence Bounds

It is possible to obtain one-sided confidence bounds

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for μ by setting either $L = -\infty$ or $U = \infty$ & replacing $z_{\alpha/2}$ by z_α.

A $100(1-\alpha)$ % upper confidence bound for μ is

$$\mu \leq U = \bar{x} + z_\alpha \sigma/\sqrt{n}.$$

and $100(1-\alpha)$ % lower confidence bound for μ is

$$\bar{x} - z_\alpha \sigma/\sqrt{n} = l \leq \mu.$$

For the sample (5, 2, 3, 1, 0, 0, 1, 3, 3) from a population,

(i) Construct a 90% CI on population mean if the popln std. devn is known to be 2.

(ii) What should be the sample size in order to keep the width of CI to less than 2.

$100(1-\alpha)$% CI

$$\mu \in \bar{x} \pm z_{0.05} * \frac{\sigma}{\sqrt{n}}$$

$$\bar{x} = \frac{5+2+3+1+0+0+1+3+3}{9} = 18/9 = 2.$$

$$CI = 2 \pm 1.645 * \frac{2}{\sqrt{9}} = 2 + 1.645 * \frac{2}{3} = 3.0967$$

$$= 2 - 1.645 * 2/3 = 0.9033.$$

$$\boxed{0.9033 \leq \mu \leq 3.0967}$$

$$\text{width} = 2 * z_{0.05} * \frac{\sigma}{\sqrt{n}} = 2 * 1.645 * \frac{2}{\sqrt{n}} < 2$$

$$= 2 * 1.645 * 2 < 2\sqrt{n}.$$

$$\text{or } \sqrt{n} > \frac{2 * 1.645 * 2}{2} = 3.29$$

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$\Rightarrow n \geq 10.824$

$\Rightarrow \boxed{n \geq 11}$

2. For the sample (5, 2, 3, 1, 4, 2, 1, 3, 3, 5) from a popul-ation representing extent of overfill in sealed beverage bottles in ml,

a) Construct a 95% confidence upper bound on population mean if the population variance is known to be 5. What should be the range of sample size in order to keep the width of the confidence interval to less than 1ml?

b) If the population variance is unknown, construct a 95% CI on the population variance.

95% upper bound on mean μ:

$$\mu \leq \bar{x} + z_{0.05} \frac{5}{\sqrt{n}}$$

$$\bar{x} = \frac{5+2+3+1+4+2+1+3+3+5}{10}$$

$$= 29/10 = 2.9$$

$$\mu \leq 2.9 + z_{0.05} * \frac{\sqrt{5}}{\sqrt{10}}$$

$z_{0.05} = 1.645$.

$$\mu \leq 2.9 + 1.645 * \sqrt{\frac{5}{10}}$$

$$\mu \leq 4.063$$

Upper bound is same as upper limit of 90% of 2 sided CI

$$= \bar{x} \pm z_{0.05} \frac{5}{\sqrt{n}}$$

$$= 2.9 \pm 1.645 \sqrt{\frac{5}{10}}$$

Width $= 2 * 1.645 \sqrt{\frac{5}{n}}$ given < 1.

$\Rightarrow 2 * 1.645 \sqrt{5} < \sqrt{n}$.

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or $n \geq (2 * 1.645)^2 * 5$

$n \geq 54.1$

∴ sample size $n \geq 54$ to keep CI width < 1 ml.

2) population variance unknown, 95% CI on popln variance.

95% CI on σ^2 : $\alpha = 0.05$ $n = 10$

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}}$$

$$\Rightarrow \frac{9 * 1.449^2}{19.02} \leq \sigma^2 \leq \frac{9 * 1.449^2}{2.7}$$

$$\Rightarrow 0.9937 \leq \sigma^2 < 7 \text{ ml}^2.$$

$s^2 = (5-2.9)^2 + (2-2.9)^2 + (3-2.9)^2 + (1-2.9)^2 + (4-2.9)^2 + (2-2.9)^2 + (1-2.9)^2 + (3-2.9)^2 + (5-2.9)^2 / 4.$

3. From a population with known standard deviation of 1.65, a sample of 32 items resulted in 34.8 as an estimate of the mean.

i) Find the standard error of the mean.

ii) Compute an interval estimate that should include the population mean 99.7 percent of the time.

i: Std error of the mean = $\sigma/\sqrt{n} = 1.65/\sqrt{32} = 0.29$

ii. $\bar{x} \pm z_{\frac{0.003}{2}} * \frac{\sigma}{\sqrt{n}} = 34.8 \pm z_{0.0015} * \frac{1.65}{\sqrt{32}}$

$= 34.8 \pm 2.97 * \frac{1.65}{\sqrt{32}}$.

$33.93 \leq \mu \leq 35.67.$

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4. The manager of cardinal Electricals light bulb division must estimate the average number of hours that a light bulb made by each bulb m/c will last. A sample of 40 light bulbs was selected from machine & the average burning time was 1416 hours and the std deviation of burning time is known to be 30 hours. Construct a 90% CI for the true population mean.

$n = 40, \quad \bar{x} = 1416 \quad \sigma = 30 \text{ hrs} \quad \alpha = 0.1$

$$C.I = \bar{X} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

$$\text{or } \bar{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1416 \pm 1.645 * \frac{30}{\sqrt{40}}$$

$$= (1408.2, 1423.8$$

$$1408.2 \leq \mu \leq 1423.8$$

5. An operating system for a PC has been studied extensively and it is known that the standard deviation of the response time foll. a particular command is $\sigma = 8$ millisecs. A new version of the operating system is installed, and we wish to estimate the mean response time for the new system to ensure that a 95% CI for μ has length at most 5 millisecs.

a) If we can assume that response time is normally distributed and that $\sigma = 8$ for the new system, what sample size would you recommend?

b) Suppose that we are told by vendor that std. devn of response time of new system is smaller say $\sigma = 6$; give the sample size that you recommend & comment on the effect the smaller std. devn has on this calculation.

$$n = \frac{(Z_{\alpha/2}\sigma)^2}{E} \quad | \quad 2E \text{ is length of CI.}$$

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a) $\sigma = 8$ length of CI = 5 CL = 95% $\alpha = 0.05$.

$$\text{length} = 2 * z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = 2 * z_{\frac{0.05}{2}} * \frac{8}{\sqrt{n}} = 5.$$

$$= 2 * 1.96 * \frac{8}{\sqrt{n}} = 5$$

$$\therefore \sqrt{n} = \frac{2 * 1.96 * 8}{5} = 6.272.$$

$$\therefore n = (6.272)^2 = 39.338.$$

$\therefore$ Sample size $n = 40$

b) std devn is $\sigma = 6$.

$$\text{i.e } 2 * 1.96 * \frac{6}{\sqrt{n}} = 5.$$

$$\therefore \sqrt{n} = 4.704 \Rightarrow n = (4.704)^2$$

$$n = 22.13 \Rightarrow n = 23.$$

We can observe that as std devn reduced the sample size also reduces.

6. A civil engineer is analysing the compressive strength of concrete. Compressive strength is normally distributed with $\sigma^2 = 1000 (psi)^2$. A random sample of 12 specimens has a mean $\bar{X} = 3250$ psi.

i) Construct a 95% two sided confidence interval on mean compressive strength.

ii) Construct a 99% two sided confidence interval on mean compressive strength.

$\sigma^2 = 1000 \text{ psi}^2$ $n = 12$ $\bar{X} = 3250$ psi

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(i) 95% CI $\alpha = 0.05$ $\alpha/2 = 0.025$

$$\mu \in \bar{X} \pm z_{0.025} * \frac{\sigma}{\sqrt{n}}$$

$$\bar{X} \pm 1.96 * \frac{\sigma}{\sqrt{n}}$$

$$\therefore CI \rightarrow 3250 - 1.96 * \frac{\sqrt{1000}}{\sqrt{12}} \leq \mu \leq 3250 + 1.96 * \frac{\sqrt{1000}}{\sqrt{12}}$$

$$\boxed{3232.1 \leq \mu \leq 3267.89}$$

(2) 99% CI $\alpha = 0.01$ $\alpha/2 = 0.005$

$$z_{\alpha/2} = 2.58$$

$$CI \rightarrow 3250 - 2.58 * \frac{\sqrt{1000}}{\sqrt{12}} \leq \mu \leq 3250 + 2.58 * \frac{\sqrt{1000}}{\sqrt{12}}$$

$$\boxed{3226.4 \leq \mu \leq 3273.55}$$

7. A random sample has been taken from a normal popula and 2 confidence intervals constructed using exactly the same data. The two CIs are (38.02, 61.98) and (39.95, 60.05)

a) What is the value of the sample mean?

b) One of these intervals is 90% CI & other is a 95% CI. Which one is the 95% CI & why?

The CI is given by $\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

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$\bar{x}$ is the midpoint of CI.

We can use either of the intervals (38.02, 61.98) or (39.95, 60.05)

$$\therefore \bar{X} = \frac{38.02 + 61.98}{2} = \frac{100.00}{2} = 50$$

$$\boxed{\bar{X} = 50}$$

b) Generally for a fixed sample size 'n' and std. devn 'σ' the higher the CL, the lower the resulting confidence Interval. We have the intervals:

(39.95, 60.05) → length or width = 20.1 — 90% CI
(38.02, 61.98) → length — 23.96 — 95% CI

Confidence Interval on the mean of a Normal Distribution, Variance unknown:

When the sample is small (<30 or <40) and σ^2 is unknown, we must make an assumption about the form of the underlying distribution to obtain a valid CI procedure. A reasonable assumption is that the underlying distn is normal.

Suppose that the population of interest has a normal distn with unknown mean μ and unknown variance σ^2. Assume that a random sample of size n, say $X_1, X_2 \cdots X_n$ is available, & let $\bar{X}$ and S^2 be sample mean & variance resp.

To construct a two sided CI on μ. If variance is known we know that $Z = (\bar{X} - \mu) / \sigma/\sqrt{n}$ has a std. normal distn. When σ^2 is unknown, a logical procedure is to replace σ with the sample std. devn: S. The random variable Z now becomes $T = (\bar{X} - \mu) / S/\sqrt{n}$.

(If n is large, proceed to use CI based on normal distn itself

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t distribution

Let $X_1, X_2, \ldots X_n$ be a random sample from a normal distn with unknown mean μ & unknown variance σ^2. The random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has 't' distribution with n-1 degrees of freedom

The t pdf is

$$f(x) = \frac{\Gamma[(k+1)/2]}{\sqrt{\pi k}\,\Gamma(k/2)} \cdot \frac{1}{[(x^2/k)+1]^{(k+1)/2}} \qquad -\infty < x < \infty$$

where k is the number of degrees of freedom.

The mean & variance of the t distribution are zero and $k/(k-2)$ for $k>2$ respectively.

The gen. appearance of the t distn is similar to the std. normal distn in that both distns are symmetric & unimodal, & the max ordinate value is reached when mean $\mu = 0$. However the 't' distn. has heavier tails than the normal; i.e it has more prob. in the tails than the normal distn. As the no of dof $k \to \infty$, the limiting form of t distn is the std. normal distn. Gen. the no. of dof for t are the no of dof associated with estimated std. dev.

Table provides percentage points of the 't' distn. We will let $t_{\alpha,k}$ be the value of random variable T with k dof above which we find an area (or prob) α. Thus $t_{\alpha,k}$ is an upper tail 100α % point of 't' distn with k dof.

For eg t value with 10 dof having an area of 0.05 to the right is $t_{0.05,10} = 1.812$

i.e $P(T_{10} > t_{0.05,10}) = P(T_{10} > 1.812) = 0.05$

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Since the t distn is symmetric about zero, we have $t_{1-\alpha,n} = -t_{\alpha,n}$; i.e t-value having an area of $1-\alpha$ to the right (& i.e an area of α to the left) is = the -ve of the t-value that has area α in the right tail of the distn.

$\therefore t_{0.95,10} = -t_{0.05,10} = -1.812$

Confidence Interval on the mean, variance unknown

If $\bar{x}$ and s are the mean and standard deviation of a random sample from a normal distn with unknown variance σ^2, a $100(1-\alpha)$ % confidence interval on μ is given by

$$\bar{x} - t_{\alpha/2,n-1}\, s/\sqrt{n} \le \mu \le \bar{x} + t_{\alpha/2,n-1}\, s/\sqrt{n}$$

where $t_{\alpha/2,n-1}$ is the upper $100\alpha/2$ % point of the t distribution with $n-1$ dof.

One sided confidence bounds on the mean of a normal distn are also of interest and can be done by using only the appropriate lower or upper confidence limit from above eqn & replacing $t_{\alpha/2,n-1}$ by $t_{\alpha,n-1}$.

It is not as easy to select a sample size 'n' to obtain a specified length (or pr. of estimation) for this CI as it was in σ known case because the length of interval involves s, n, & $t_{\alpha/2,n-1}$. Note that t-percentile depends on sample size n. Consequently an appropriate n can only be obtained through trial & error.

Problems:

Det the t-percentile that is required to construct each of the following.

a) Two sided CI: confidence level = 95%. dof = 12

$t_{\alpha/2,n-1} = t_{0.05/2,12} = 2.179$.
(dof)

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ii) Confidence level = 99.9% ∴ dof = 15.

$t_{\alpha/2, n-1} = t_{0.001/2, 15} = 4.073$

b. One sided CI.

(i) Confidence level = 99%. dof = 19

$t_{\alpha, n-1} = t_{0.01, 19} = 2.539$

(ii) CL = 99.9% dof = 24

$t_{\alpha, n-1} = t_{0.001, 24} = 3.467$ if dof = 15.

$t_{0.001, 15} = 3.733$.

2. A research engineer for a tire manufacturer is investigating tire life for a new rubber compound & has built 16 tires & tested them to end of life in a road test. The sample mean & std. devn are 60,139.7 & 3645.94 kms. Find a 95% CI on mean tire life.

μ - mean tire life

Soln

$$C.I = \bar{X} - t_{\alpha/2, n-1}\frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2, n-1}\frac{s}{\sqrt{n}}$$

$$= 60139.7 - 2.131 * \frac{3645.94}{\sqrt{16}} \leq \mu \leq 60139.7 + 2.131 * \frac{3645.94}{\sqrt{16}}$$

$\alpha = 0.05$ $n = 16$.

$t_{\alpha/2, n-1} = t_{0.025, 15} = 2.131$

$$= [58197.33, 62082.07].$$

3. In a maintenance department of a firm, the life of the electronic equipment is found to be normally distributed. A random sample of 30 sensors has a sample mean life of 800 hours, and a sample std. devn. of 70 hours.

(i) Construct a 95% two sided CI for the mean life of the sensors.

(ii) Construct a 99% two sided CI for the variance of life of sensors.

(iii) If the popln std. devn of life of sensors is known

to be 90 hours, construct a 99% 2 sided CI for mean life of the sensors.

iv) Comment on difference in answers to (i) & (iii).

Life $\sim N(\mu, \sigma^2)$, $\bar{x} = 800$ hrs $s = 70$ hrs. $\alpha = 0.05$ $n = 30$

$t_{\alpha/2, n-1} = t_{0.025, 29} = 2.045$

(i) $CI = \bar{X} - t_{\alpha/2} * \frac{s}{\sqrt{n}} \leq L \leq \bar{X} + t_{\alpha/2} \frac{s}{\sqrt{n}}$

$= 800 - 2.045 * \frac{70}{\sqrt{30}} \leq L \leq 800 + 2.045 * \frac{70}{\sqrt{30}}$

$= 800 \pm 26.135$

$= [773.86, 826.14]$ hrs

(ii) CI on variance.

$\sigma^2 \in \frac{(n-1)s^2}{\chi^2_{0.01/2, n-1}}$ and $\frac{(n-1)s^2}{\chi^2_{1-\frac{0.01}{2}, n-1}}$

$= \frac{29 * 70^2}{52.34}$ & $\frac{29 * 70^2}{13.12} = [2715, 10,830.7]$

iii) $CI = \bar{x} \pm Z_{\frac{0.01}{2}} \frac{\sigma}{\sqrt{n}}$

$= 800 \pm 2.58 * \frac{90}{\sqrt{30}} = 800 \pm 42.39$

$= [757.6, 842.4]$

Confidence Interval of (iii) is wider ∵ of 99% CI in spite of population variance being known.

4. The life of an electronic sensor is normally distributed. A random sample of 20 sensors has a mean life of 1000 hrs, and a standard deviation of 100 hours.

(i) Construct a 99% 2 sided CI for the mean life of the sensors.

(ii) Construct a 95% 2 sided CI for the variance of life of sensors.

iii If the popln std devn of the life of sensors is known to be 105 hours, construct a 99% 2 sided CI for mean life of the sensors.

iv Comment on diff in answers to (ii) & (iii)

Soln $LNN(\mu, \sigma^2)$ $\bar{x} = 1000$ hrs $s = 100$ hrs, $n = 20$

(i) $\mu \in \bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$.

99% CL → 2 sided ; $\alpha = 1\% = 0.01$

$t_{\alpha/2, n-1} = t_{\frac{0.01}{2}, 19} = t_{0.005, 19} = 2.861$

$CI = \bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$

$= 1000 \pm \frac{2.861 * 100}{\sqrt{20}}$ $= 936.03 \leq \mu \leq 1063.97$

$= [936.03, 1063.97]$ hrs.

(ii) $\sigma^2 \in \frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}}$ $\alpha = 0.05$. $\alpha/2 = 0.025$.

$= \frac{19 * 100^2}{32.85}, \frac{19 * 100^2}{8.91}$ $\chi^2_{0.025, 19} = 32.85$

$\chi^2_{(1-0.025), 19} = \chi^2_{0.975, 19} = 8.91$

$= (5783.87, 21324.35)$ → (variance).

(iii) $\mu \in \bar{x} + Z_{\frac{0.01}{2}} \frac{\sigma}{\sqrt{n}}$. $\sigma = 105$. $\alpha = 0.01$

$\alpha/2 = 0.005$

$= 1000 + 2.58 * \frac{105}{\sqrt{20}}$.

$= (939.42, 1060.58)$ hrs.

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(iv) CI of (iii) is narrower as compared to (i) since σ is known. A more precise interval is obtained when additional information is available.

5. For the sample (5, 2, 3, 1, 0, 0, 1, 3, 3) from a popln Construct a 95% CI on popln mean if the popln variance is unknown.

$$\mu \in \bar{x} \pm t_{\frac{0.05}{2}, n-1} \frac{s}{\sqrt{n}}$$

$$2 \pm t_{0.025, 8} * \frac{1.658}{\sqrt{9}}.$$

$$2 \pm 2.306 * \frac{1.658}{\sqrt{9}}.$$

$$CI = [0.7253, 3.2747].$$

$\bar{X} = 2.$

$$S^2 = \frac{(5-2)^2 + (2-2)^2 + (3-2)^2 + (1-2)^2 + (0-2)^2 + (0-2)^2 + (1-2)^2 + (3-2)^2 + (3-2)^2}{8}$$

$$= \frac{9+0+1+1+4+4+1+1+1}{8}$$

$$= \frac{22}{8} = 2.75$$

$$S = 1.658$$

6. Cloud seeding has been studied for many decades as a weather modification. The rainfall in acre feet from 20 clouds that were selected at random & seeded with silver nitrate are as follows:

18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, & 31.6

(a) Construct a 95% one sided lower bound on the mean diameter.

(b) Construct a 90% 2sided CI on mean dia. What is the width of this interval?

c) If the width of interval in part (b) were to be halved, how much should be the sample size? Use trial & error method if needed.

a) 95% lower bound: $\bar{x} - t_{0.05, 19} * \frac{S}{\sqrt{n}}$

$n = 20, \quad S = 4.7848 \quad \bar{x} = 26.035$

$t_{0.05, 19} = 1.729.$

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$\Rightarrow$ lower bound $\mu \geq 26.035$ $\frac{1.7343}{\sqrt{20}}$

$\boxed{\mu \geq 24.185}$

b) 90% CI $\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$ $\alpha = 0.10$ $\alpha/2 = 0.05$ $t_{0.05, 19} = 1.729$

$26.035 \pm 1.729 * \frac{4.7848}{\sqrt{20}}$

90% CI = (24.185, 27.885)

width = max − min = 3.7958

(c) width $= 2 t_{0.05, n-1} s/\sqrt{n}$ $n=30$ $\alpha = 0.1$ $\alpha/2 = 0.$ [Assuming s remains

$n = 30 \Rightarrow$ width $= 2 * 1.697 * \frac{4.7848}{\sqrt{30}} = 2.96$

$n = 40 \Rightarrow$ width $= 2 * 1.684 * (4.7848/\sqrt{40}) = 2.5$

$n = 60 \Rightarrow$ width $= 2 * 1.671 * 4.7848/\sqrt{60} = 2.06$

$n = 120 \Rightarrow$ width $= 2 * 1.658 * 4.7848/\sqrt{120} = 1.4484$

$\frac{1}{2}$ width of part(b) is $3.7958/2 = 1.8979$

$\boxed{\therefore n \text{ should be between } 60 \text{ \& } 120}$

7 (a) From a sample of 42 gasoline stations statewide, the avg price of a gallon of unleaded gas was found to be \$1.12 & std. devn. was \$0.04/gallon. Construct a 99% CI on mean price of a gallon of unleaded gas

b) A random sample of 51 vehicles on a highway clocked speeds with an avg of 23.2 mph & std. devn of 0.3 mph.

(i) Estimate the std. devn of the population.

(ii) " " std. error of the mean for this popln

(a) $n=42, \quad \bar{x}=1.12 \quad s=0.04 \quad \alpha=0.01$

$$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} = 1.12 \pm 2.7 * \frac{0.04}{\sqrt{42}}$$

$$t_{\frac{0.01}{2}, 41} = t_{0.005, 41} = 2.7$$

99% CI = [\$1.10, \$1.14]

b. $n=57 \quad \bar{x}=23.2 \quad s=0.3$ assume $\alpha=0.05$

(i) $\left(\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2}, n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \right) = \frac{56*0.3^2}{77.4}, \frac{56*0.3^2}{36.5}$

95% CI for $\sigma^2 = (0.065, 0.138) \Rightarrow$

CI for $\sigma = (.255, 0.37)$

(ii) Std error of mean $= \frac{(.255+0.37)}{2} * \frac{1}{\sqrt{57}}$

$= 0.041$

8. Level of contamination in % was studied in 6 packs of food with following data:
16.8, 17.2, 17.4, 16.9, 16.5, 17.1. Find a 99% confidence interval on the mean % μ. & provide a practical interpretation of this interval.

$\bar{x} = 101.9/6 = 16.98.$

$$s^2 = \frac{0.0324 + 0.0484 + 0.1764 + 0.006 + 0.2304 + 0.0144}{5}$$

$s^2 = 0.1016.$

$s = 0.318.$

$$99\ CI = \bar{X} \pm t_{\frac{0.01}{2}, 5} \frac{S}{\sqrt{n}}$$

$$= 16.98 \pm 4.032 * \frac{0.318}{\sqrt{6}}$$

$$= 16.98 \pm 0.523$$

$$= [16.457, 17.503]$$

9. The wall thickness of 25 glass bottles was measured & the sample mean was 4.05mm & sample std devn was 0.08mm. Find a 95% lower confidence bound for mean wall thickness, & interpret the interval.

$n = 25 \quad \bar{X} = 4.05 \quad S = 0.08 \quad \alpha = 0.05$

$$\bar{X} - t_{\alpha, n-1} * \frac{S}{\sqrt{n}} \leq \mu \qquad t_{0.05, 24} = 1.711$$

$$\mu \geq \bar{X} - t_{\alpha, n-1} * \frac{S}{\sqrt{n}}$$

$$\mu \geq 4.05 - 1.711 * \frac{0.08}{\sqrt{25}}$$

$$\mu \geq 4.02mm$$

Min Population mean wall thickness is 4.02mm

CONFIDENCE INTERVAL ON THE VARIANCE AND STANDARD DEVIATION OF A NORMAL DISTRIBUTION

χ^2 Distribution.

Let $X_1, X_2 \ldots X_n$ be a random sample from a normal distribution with mean μ and variance σ^2, and let S^2 be the sample variance. Then the random variable.

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}.$$

has a chi square (χ^2) distribution with $n-1$ dof.

Confidence Interval on the variance.

If s^2 is the sample variance from a random sample of n observations from a normal distn with unknown variance σ^2, then a $100(1-\alpha)\%$ confidence interval on σ^2 is

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}$$

where $\chi^2_{\alpha/2,n-1}$ & $\chi^2_{1-\frac{\alpha}{2},n-1}$ are the upper & lower $100\alpha/2\%$ pts of chi-square distn with $n-1$ dof resp.

A confidence interval for σ has lower & upper limits that are square roots of corresponding limits

One sided Confidence Bounds on the variance

The $100(1-\alpha)\%$ lower & upper confidence bounds on σ^2 are

$$\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} \leq \sigma^2 \quad \text{and} \quad \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{(1-\alpha),n-1}}$$

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Problems.

1. Determine the values of the following percentiles.

a) $\chi^2_{0.05,10} = 18.31$.
b) $\chi^2_{0.995,16} = 5.14$.
c) $\chi^2_{0.005,25} = 46.93$.

2. Det the χ^2 percentile that is required to construct each of the foll. CIs.

(a) Confidence level = 95%. dof = 24 one sided ~~upper~~ lower.

$$\chi^2_{0.05,24} = 36.42.$$

(b) CL = 99%. dof = 9 one sided ~~lower~~ upper.

$$\chi^2_{1-0.01,9} = \chi^2_{0.99,9} = 2.09.$$

(c) CL = 90%. dof = 19 two sided

$$\chi^2_{0.1/2,19} = \chi^2_{0.05,19} = \underline{30.14}$$

2. A rivet is to be inserted into a hole. A random sample of n = 15 parts is selected, and the hole diameter is measured. The sample std. devn. of the hole diameter measurements is s = 0.008 mm. Construct a 99% lower confidence bound for variance.

$n = 15 \qquad s = 0.008$

Lower confidence bound $\frac{(n-1)s^2}{\chi^2_{0.01,n-1}} \leq \sigma^2$.

$$\chi^2_{0.01,15-1} = \chi^2_{0.99,14} = \text{~~4.665~~} \; 29.14$$

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$$\sigma^2 \geq \frac{14 * 0.008^2}{29.14}$$

$$\sigma^2 \geq 3.074 * 10^{-5}$$

$$CI = [0.00003074, \infty]$$

The

3 A sugar content of the syrup in canned peaches is normally distributed. A random sample of $n=10$ cans yields a sample std. devn of $s=4.8mg$. Calculate a 95% two sided CI for σ.

$n=10$ $s=4.8mg$ $CL=95\%$ $\alpha=0.05$.

$$\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2},n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{(1-\frac{\alpha}{2}),n-1}}$$

$\chi^2_{0.025,9} = 19.02$

$$\frac{9*4.8^2}{19.02} \leq \sigma^2 \leq \frac{9*4.8^2}{2.70}$$

111th $\chi^2_{1-0.025,9} = \chi^2_{0.975,9} = 2.70$

$$10.90 \leq \sigma^2 \leq 76.8.$$

$$\therefore \sqrt{10.9} \leq \sigma \leq \sqrt{76.8} \Rightarrow 3.30 \leq \sigma \leq 8.76$$

4. For the sample (5, 2, 3, 1, 0, 0, 1, 3, 3) from a population,
(i) Construct a 95% confidence interval on population variance.

$s=1.6583$. $n=9$

$$\frac{(n-1)s^2}{\chi^2_{0.05,8}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{0.95,8}}$$

$$\frac{8*1.6583^2}{15.51} \leq \sigma^2 \leq \frac{8*1.6583^2}{2.73} \Rightarrow 1.4189 \leq \sigma^2 \leq 8.058$$

$\Rightarrow \boxed{1.19 \leq \sigma \leq 2.838}$

5. Amino acid concentrations in 8 packs of dog feeds were recorded in mg/kg of dog feed as follows:
22, 24, 20, 26, 27, 24, 26, 23.

(i) Perform a test at 5% significance to see if the population variance is greater that $5.5 (mg/kg)^2$ What is the p-value of the test?

(ii) Construct an upper and a lower 90% confidence bounds for the population variance.

Soln $H_0: \sigma^2 = 5.5$. $H_1: \sigma^2 > 5.5 (mg/kg)^2$

$\alpha = 0.05$ $n = 8$. $\bar{x} = 24$ $\boxed{s = 2.3299}$

$$s^2 = \frac{4+0+16+4+9+0+4+1}{7} = \frac{38}{7} = 5.428$$

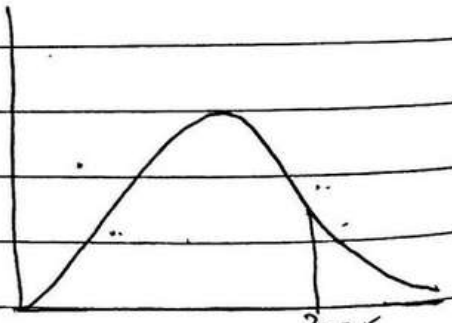
$$s = \sqrt{5.428} = 2.3299$$

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{7 * 2.3299^2}{5.5} = 6.9091$$

~~1.256~~

$\chi^2_{tab} = \chi^2_{0.05,7}$ $\chi^2_{0.05,7} = 14.07$

Do not reject H_0.



P-value = $P(\chi^2_7 > 6.91) =$ between .4 and .5

Since $P(\chi^2_7 > 6.346) = .5$ and $P(\chi^2_7 > 7.283) = 0.4$

b. 90% upper bound : $\sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha, n-1}}$

$$\Rightarrow \sigma^2 \leq \frac{7 * 5.4286}{2.833} = 13.41.$$

90% of lower bound: $\frac{(n-1)s^2}{\chi^2_{\alpha, n-1}} \leq \sigma^2$.

$$\Rightarrow \sigma^2 \geq \frac{7 * 5.4286}{12.02} = 3.16.$$

i.e $3.16 \leq \sigma^2 \leq 13.41$.

Question No 3. continued.

a) Find a 90% 2 sided confidence interval for population standard deviation.

b) Find a 95% upper bound and a 95% lower bound for population standard deviation. How do these bounds compare with the limits of part (a)?

$n = 10$ $s = 4.8$

a) $\alpha = \cancel{0.01}\ 0.1$ 2sided $= \alpha/2 = \cancel{0.005}\ 0.05$.

90% of CI for variance $= \left[\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}}\right]$

$\chi^2_{0.05, 9} = 16.92$.

$\chi^2_{0.95, 9} = 3.325$.

Interval for σ^2: $\left[\frac{9 * 4.8^2}{16.92}, \frac{9 * 4.8^2}{3.325}\right] = [12.26, 62.36]$

Interval for σ: $(3.5, 7.9)$.

b. 95% upper bound $\alpha = 0.05$. $1 - \alpha = 0.95$.

$\chi^2_{0.95, 9} = 3.325$

$\sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha, n-1}}$ $\sigma^2 \leq \frac{9 * 4.8^2}{3.325}$

$\sigma^2 \leq 62.36 \Rightarrow \boxed{\sigma \leq 7.9}$

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95% lower bound : $\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{\alpha, n-1}} = \frac{9 * 4.8^2}{16.92}$

$\sigma^2 \geq 12.26 \Rightarrow \boxed{\sigma > 3.5}$

Both bounds are two-sided limits of two sided CI since $\alpha/2$ in part A is same as α in part B.

7. A hole dia is measured in random sample of 15 parts. The sample standard deviation of the hole dia is 0.008 mm. Construct a 99% lower confidence bound for σ^2.

$n = 15 \quad s = 0.008.$

99% CI $\Rightarrow \alpha = 0.01$

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{\alpha, n-1}} = \frac{(14)\ 0.008^2}{\chi^2_{0.01, 14}}$$

$$= \frac{14 * 0.008^2}{29.14} = 3.84 * 10^{-3}$$

$$\sigma^2 \geq 3.84 * 10^{-3} \Rightarrow \boxed{\sigma > 0.062}$$

8. A set of 15 wt. measurements give sample std. deviation $s = 0.0089$ gm.

i. Does the population std. devn σ of all the weight measurements exceed 0.008 gms at 5% significance level? Find the P-value for this test.

ii) Answer part (a) with the help of a lower confidence interval for σ with $\alpha = 5\%$.

$n = 15 \quad s = 0.0089$ gm.

(i) $H_0: \sigma = 0.0089$ gm $\quad H_1: \sigma > 0.0089$ gm.

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$\alpha = 0.05$.

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{14 * 0.0088^2}{0.008^2} = 16.94$$

$\boxed{\chi_0^2 = 16.94}$

[pvalue → 16.94 lies b/w 0.5 & 0.1]

$\chi^2_{\text{from tables}} = \chi^2_{0.05,14} = 23.68$ [$0.1 < p < 0.5$ · $21.06 < p < 13.34$]

Since $\chi_0^2 < \chi^2_{table}$ we fail to reject null hypothesis or std devn does not exceed 0.008gm at 5% α.

ii lower confidence bound or interval.

$\alpha = 0.05$.

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{0.05,14}}$$

$$\sigma^2 \geq \frac{14 * 0.0088^2}{23.68} = 4.578 * 10^{-5}.$$

$$\Rightarrow \boxed{\sigma > 6.76 * 10^{-3} \doteq 0.0067}$$

9. The % of titanium in an alloy used in aerospace casting is measured in 51 randomly selected parts. The sample standard deviation is $s = 0.28$. Construct a 95% two-sided confidence interval for σ.

95% CI for σ given $n = 51$ $s = 0.28$.

$\alpha = 0.05$, $n = 51$ $\chi^2_{\alpha/2,n-1} = \chi^2_{0.025,50} = 71.42$.

$$\chi^2_{1-\alpha/2,n-1} = \chi^2_{0.975,50} = 32.36.$$

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{(1-\frac{\alpha}{2}),n-1}}$$

$$\frac{50*(0.28)^2}{71.42} \leq \sigma^2 \leq \frac{50*(0.28)^2}{32.36}$$

$$0.055 \leq \sigma^2 \leq 0.121$$

Take the square root of the endpoints of this interval to obtain.

$$\boxed{0.234 < \sigma < 0.348}$$